Pre-Interview Case: Factor Analysis for a Systematic Long/Short Equity Portfolio

Background

Presto runs fully systematic long/short equity strategies across multiple global markets.

The strategies generally have the following characteristics:

- Short holding periods, typically from a few hours to a few days
- High turnover
- Broadly diversified stock universes
- Long/short and generally low directional market exposure
- Systematic portfolio construction and algorithmic execution
- Machine-learning-based prediction models, including tree-based and deep-learning models
- Some models are therefore relatively “black-box” from a risk-management perspective
- The objective is to generate high risk-adjusted returns

We are interested in building a practical factor-analysis framework for Middle Office / Risk Management. The purpose is not to develop alpha signals, but to better understand portfolio risk, performance drivers, diversification, and common exposures across strategies and trading teams.

Discussion Question

Assume that you are the risk manager responsible for designing this framework from scratch.

Please think about the following questions.

1. How can a factor framework help the firm?

How would factor analysis provide useful information to:

- Risk Management / Middle Office
- Quant Traders / Portfolio Managers
- Senior Management / Capital Allocation

What decisions or questions should the framework help these stakeholders address?

2. What factors would you consider relevant?

Given the characteristics of the strategies above, what types of factors would you include in the framework?

Please explain:

- why you selected them
- which traditional equity factors you would consider less relevant
- whether you would use different factors for different markets, horizons, or strategies

3. How would you implement the framework?

If you were building the framework from scratch, how would you approach the design and implementation?

Please discuss the main methodological choices, data requirements, estimation issues, validation approach, and how you would present the outputs to users.

Interview Format

There is no single correct answer.

We are primarily interested in your reasoning, practical experience, prioritization, and judgment rather than a textbook description of factor investing.

Please think through the problem in advance. During the interview, we will ask you to explain your proposed framework, the trade-offs involved, and how you would determine whether it is useful in practice.

No coding or formal written submission is required.